Course: Artificial Intelligence	**Assignment 1**	DUE DATE: 15/09/2026
Department: BCY/BSE		Total Questions: 4 Total Marks: 100

Instructions:

- **Attempt question 1-3 by hand.**
- **Scan your handwritten work and submit it as a PDF on GCS.**
- **Submit Question 4 Source code files separately (any language, preferably Python).**
- **No plagiarism—copied work gets zero.**

Q.1 An autonomous vehicle is travelling at high speed when its sensors detect an unexpected obstacle. The vehicle must make a decision within milliseconds.The vehicle uses cameras, LiDAR, machine-learning models, sensor fusion, and a decision-making module. It can either:

A. continue forward and potentially hit the obstacle,
B. suddenly change lanes and potentially collide with another vehicle,
C. brake sharply, potentially injuring passengers,
D. take another action based on its learned policy.

Answer the following:

1) Explain how an intelligent agent would perceive, reason about, and act upon this situation. Identify the PEAS components of the autonomous-driving agent.
2) Draw and explain an architecture showing sensing, perception, state estimation, prediction, planning, decision-making, and action.
3) Suppose the AI makes a decision that maximizes the expected number of people saved but results in the death of one passenger. Is this an acceptable AI decision? Discuss the issue from the perspectives of rationality, utility, safety, ethics, and responsibility. There is no single correct yes/no answer; marks will be awarded based on the quality of the justification.
4) The AI performs extremely well in normal driving conditions but fails when encountering an unusual situation that was absent from its training data. What does this reveal about the difference between learning and intelligence?
5) Identify the major sources of uncertainty in this system and explain how uncertainty can affect its reliability.

Q.2 A hospital uses an autonomous mobile robot to deliver medicines and laboratory samples between different departments. The hospital consists of several connected corridors and rooms. Some corridors

may become temporarily unavailable because of cleaning, medical emergencies, or equipment movement. The robot starts from the pharmacy and must deliver medicine to the appropriate ward. The robot has access to a digital map of the hospital, but the map may not always reflect the current situation. The robot can detect nearby obstacles using sensors, but it cannot observe the entire hospital at once. The robot can move forward, backward, left, or right, wait when necessary, and update its internal map when it detects a blocked corridor. It can carry only one delivery at a time. The hospital wants the robot to reach its destination safely and efficiently. A shorter route is desirable, but entering a dangerous or blocked area should be avoided even if doing so would produce a shorter route.

1. Identify the implicit intelligent agent in the scenario and describe its environment, sensors, actuators, and overall objective.
2. Formulate the medicine-delivery task as an AI problem by defining the state representation, initial state, actions, transition model, goal condition, path-cost function, and a possible solution. Justify why your proposed state representation contains sufficient information for decision-making.
3. The hospital administrator claims that, because the robot has access to a complete digital map, Breadth-First Search will always produce the best route. Critically evaluate this claim and provide a counterexample in which BFS finds a solution but does not produce the most desirable solution.
4. Consider two robots. Robot X always follows the shortest route calculated from its original map, while Robot Y sometimes selects a longer route after detecting unexpected obstacles because it predicts that the original route is likely to fail. Robot X produces shorter planned routes, but Robot Y completes more deliveries successfully. Which robot demonstrates more rational behaviour? Justify your answer using the concepts of rationality, uncertainty, adaptation, and performance measures.
5. For each of the following statements, state whether you agree or disagree and provide a logical justification or counterexample where appropriate:
    A. An agent with incomplete information about its environment can never make a rational decision.
    B. If an agent selects the shortest available path, its decision is necessarily rational.
    C. Two rational agents operating in the same environment may select different actions.
    D. If an action produces a bad outcome, the agent that selected that action must have behaved irrationally.
    E. An agent may rationally select an action that temporarily increases its distance from the goal.
    F. An agent that always selects the action with the lowest immediate cost will always minimise the total cost of reaching the goal.
    G. An agent can make a rational decision even when its model of the environment is incomplete.

Q.3 A university has developed an Autonomous Emergency Response Robot (AERR) to operate inside buildings during emergencies such as fires, earthquakes, and structural failures. The robot is provided with a map consisting of connected rooms and corridors. Each room is represented as a node, while each corridor represents a connection (edge) between two nodes. The robot must navigate from a source location to a target location. A team member makes the following claim: "Because the robot has a

complete map of the building, Depth-First Search (DFS) is the best search strategy. It uses very little memory, will eventually find the target, and will also find the shortest route." Critically evaluate this claim and answer the following questions.

(a) Is the above statement completely correct? Explain your answer by discussing the completeness, optimality, and memory requirements of DFS.

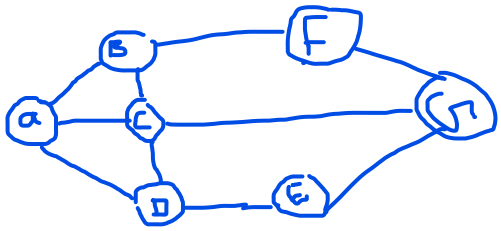
(b) Analyze whether DFS is complete and optimal under each of the following conditions:

1. Finite search tree
2. Infinite search tree
3. Graph containing cycles

For each case, clearly state whether DFS is complete and/or optimal and provide a brief justification.

(c) Construct a small search graph containing 5–7 locations that demonstrates a situation where DFS finds a valid path to the goal, but the path returned by DFS is not the shortest path. Your graph must clearly include:

- Start node
- Goal node
- All nodes and connections
- Direction/order of exploration
- DFS traversal/expansion order
- Path returned by DFS
- Actual shortest path
- Length of both paths



a → B → F → G = 3

a → C → G = 2

3 − 2 = 1

Use graph-search DFS: maintain a visited set so that a node is not expanded more than once. When multiple unexplored neighbors are available, select the leftmost neighbor first. Assume every edge has unit cost (cost = 1).

Important: Your example must be your own construction and should clearly demonstrate why DFS can return a non-optimal solution.

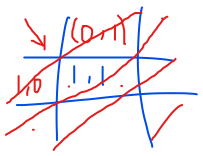
(d) Explain why having a complete map of the environment does not automatically make DFS an optimal search strategy. Your answer must clearly distinguish between:

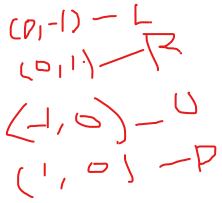
1. Knowing the search space, having information about all available locations and connections.
2. Selecting the next node to explore , the decision made by DFS.
3. Knowing the path cost, understanding the distance/cost associated with different paths.
4. Finding an optimal solution, selecting the solution with the minimum cost or minimum number of steps.

Conclude by explaining why DFS can have complete information about the environment but still return a non-optimal solution.

Q.4 The Autonomous Emergency Response Robot (AERR) is operating inside a damaged building. The building is represented as a 2D grid, where some cells are accessible and others are blocked due to debris. The robot must navigate from the starting position (S) to the goal position (G). Path Length: Path length is defined as the number of moves (edges) from the start node to the goal node. The starting node is not counted as a move. Node Expansion: A node is counted as expanded when it is removed from the frontier and its successors/neighbors are examined. If the goal is removed from the frontier and the search terminates before generating its successors, the goal is not counted as an expanded node. The robot can move only in the following four directions: Down → Right → Up → Left. Diagonal movement is not allowed. DLS Depth: For Depth-Limited Search (DLS), the starting node S is at depth 0. A path containing N moves therefore has depth N. The following grid represents the building:

```
maze = [
  ['S', 0, 1, 0, 0, 0],
  [0,   0, 1, 0, 1, 0],
  [0,   1, 0, 0, 1, 0],
  [0,   0, 0, 1, 0, 0],
  [1,   1, 0, 0, 0, 'G']
]
```





Cell Representation

Symbol	Meaning
S	Starting position
G	Goal position
0	Free/accessible cell
1	Obstacle/blocked cell

a) Search Algorithm Implementation: Implement the following search algorithms in Python:

- Breadth-First Search (BFS)
- Depth-Limited Search (DLS)
- Iterative Deepening Search (IDS)
- Your program must allow the algorithms to search for a path from S to G. For each algorithm, your program must report:
- Path found from S to G
- Path length

- Number of nodes expanded
- Whether the goal was successfully found

Implementation Constraints: You must implement the search algorithms yourself. You are not allowed to use: queue, deque, Queue, Built-in graph/search libraries, Any library that directly implements BFS, DFS, DLS, or IDS. You may use basic Python data structures and constructs, including:

- Lists
- Dictionaries
- Sets
- Tuples
- Loops
- Conditional statements
- User-defined functions

For BFS, DLS, and IDS, the main search logic must be implemented manually.

Search Implementation Requirements

1. **BFS, DLS, and IDS must use the specified neighbor order: Down → Right → Up → Left**
2. **For DLS, the starting node S is at depth 0.**
3. **For IDS, the initial depth limit must be 0.**
4. **IDS must increase the depth limit by 1 after each unsuccessful iteration until the goal is found.**
5. **For every algorithm, count nodes according to the Node Expansion definition given above.**
6. **Do not count the goal as expanded if the search terminates immediately when the goal is removed from the frontier, before its successors are generated.**

b) Execute all three algorithms on the given maze and record their results. Prepare a comparison table using the following format:

Algorithm	Path Found	Path Length	Nodes Expanded	Goal Found?
BFS				
DLS (Limit = 9)				
IDS (until goal found)				

Since DLS depends on a depth limit, execute DLS using the following depth limits: 2, 4, 6, and 9

For each depth limit, report:

- Whether the goal was found
- Path returned, if any
- Path length

- Number of nodes expanded

Depth Limit	Goal Found?	Path	Path Length	Nodes Expanded
2				
4				
6				
9				

Note: For this maze, the shortest solution has a depth of 9 moves. Therefore, depth limits of 2, 4, and 6 are insufficient to reach the goal, while a depth limit of 9 is sufficient.

IDS Experiments: For IDS, start with depth limit 0 and increase the limit by 1 after each unsuccessful iteration until the goal is found. Report the number of nodes expanded at each depth-limit iteration and the total number of nodes expanded.

IDS Depth Limit	Goal Found?	Path Length	Nodes Expanded
0			
1			
2			
3			
—			
9			

c) Based on your implementation and experimental results, answer the following:

1. Which algorithm found the shortest path in the given maze? Explain why this algorithm guarantees a shortest path under the conditions of this problem. If multiple shortest paths exist, report the path produced by your implementation.
2. Which search strategy is expected to use the least memory among BFS, DLS, and IDS? Explain your answer based on how the algorithm stores and explores nodes.
3. Why can Depth-Limited Search fail to find the goal even when a valid path exists?
4. How does Iterative Deepening Search overcome the depth-limit problem of DLS?
5. Explain how IDS changes the depth limit during the search. IDS may expand some nodes multiple times. Why does this happen, and is this repeated work a significant disadvantage?
6. Compare BFS, DLS, and IDS in terms of:
    - Time complexity

- Space complexity
- Completeness
- Optimality
- Assume that every move has equal cost (cost = 1).

7. Suppose the goal G is moved to a location that is much deeper inside the maze. Predict how the performance of the following algorithms would change:
   - BFS
   - DLS
   - IDS
8. Which algorithm would you expect to perform better, and why? Support your answer using the concepts of solution depth, branching factor, and memory usage.
9. Now assume that the maze is increased from its current size to a 100 × 100 grid. Assume the grid remains unweighted, and the robot can move only in the four cardinal directions: Down, Right, Up, and Left.
   Would your recommendation remain the same? Compare BFS, DLS, and IDS with respect to:
   - Time complexity
   - Space complexity
   - Solution depth
   - Completeness
   - Optimality
10. Conclude by identifying which strategy you would choose for a large emergency-navigation problem and justify your decision rather than simply naming an algorithm.

Your submission must contain:

1. **Complete Python source code for BFS, DLS, and IDS.**
2. **Output produced by running the algorithms.**
3. **Required comparison tables.**
4. **Answers to all critical-analysis questions.**
5. **A brief explanation of how your implementation works.**